K-medoids

Example from textbook

In this article, We'll describe the PAM algorithm and provide practical examples using **R** software. In the next chapter, we'll also discuss a variant of PAM named **CLARA** (Clustering Large Applications) which is used for analyzing large data sets.

Note that, in practice, you should get similar results most of the time, using either euclidean or Manhattan distance. If your data contains outliers, Manhattan distance should give more robust results, whereas euclidean would be influenced by unusual values.

```
data("USArrests")      # Load the data set
df <- scale(USArrests) # Scale the data
head(df, n = 3)        # View the firt 3 rows of the data
```

```
##             Murder   Assault   UrbanPop         Rape
## Alabama 1.24256408 0.7828393 -0.5209066 -0.003416473
## Alaska  0.50786248 1.1068225 -1.2117642  2.484202941
## Arizona 0.07163341 1.4788032  0.9989801  1.042878388
```

Loading Pacakges

```
library(cluster)
library(factoextra)
```

```
## Warning: package 'factoextra' was built under R version 4.0.5
```

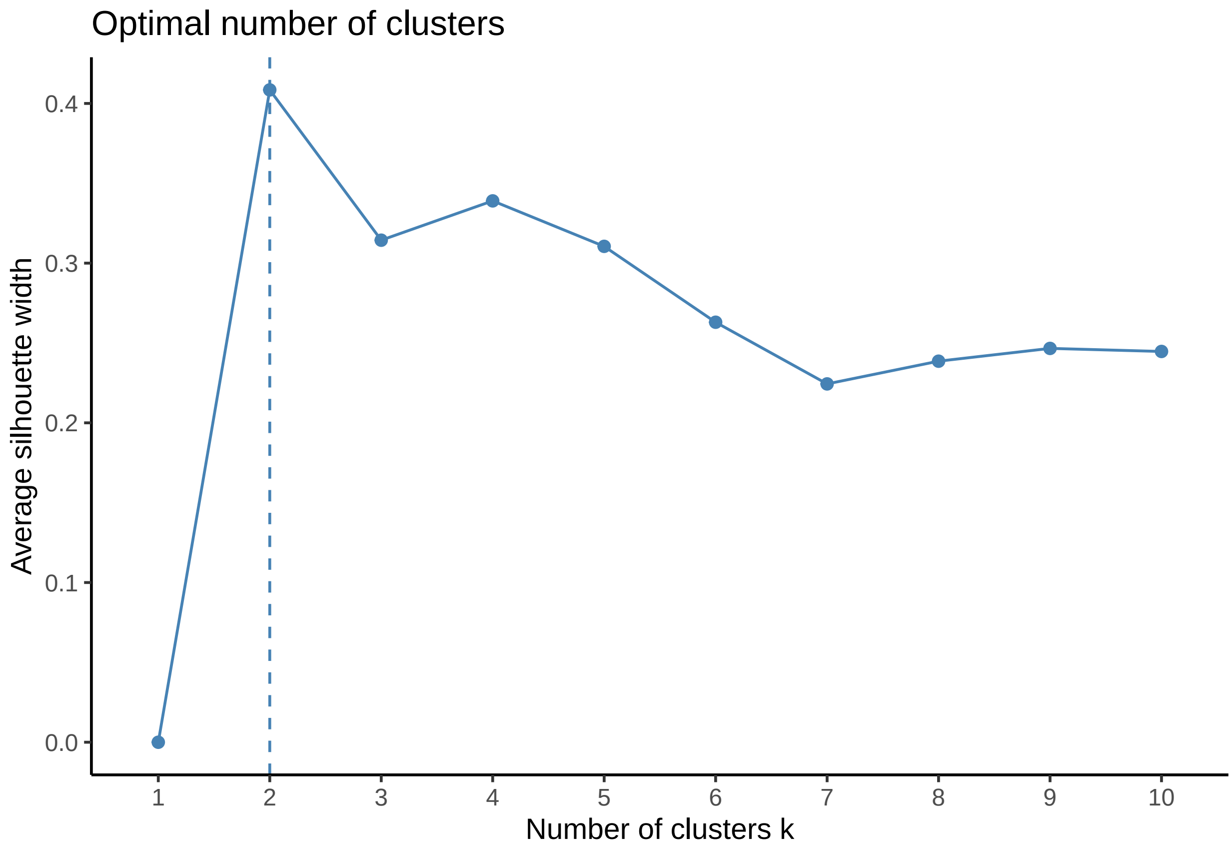
```
## Loading required package: ggplot2
```

```
## Warning: package 'ggplot2' was built under R version 4.0.5
```

```
## Welcome! Want to learn more? See two factoextra-related books at https://goo.gl/ve3WBa
```

Estimating the optimal number of clusters

```
library(cluster)
library(factoextra)
fviz_nbclust(df, pam, method = "silhouette")+
  theme_classic()
```



From the plot, the suggested number of clusters is 2. In the next section, we'll classify the observations into 2 clusters.

```
pam.res <- pam(df, 2)
print(pam.res)
```

```
## Medoids:
##            ID     Murder    Assault   UrbanPop       Rape
## New Mexico 31  0.8292944  1.3708088  0.3081225  1.1603196
## Nebraska   27 -0.8008247 -0.8250772 -0.2445636 -0.5052109
## Clustering vector:
##        Alabama         Alaska        Arizona       Arkansas     California
##              1              1              1              2              1
##       Colorado    Connecticut       Delaware        Florida        Georgia
##              1              2              2              1              1
##         Hawaii          Idaho       Illinois        Indiana           Iowa
##              2              2              1              2              2
##         Kansas       Kentucky      Louisiana          Maine       Maryland
##              2              2              1              2              1
##  Massachusetts       Michigan      Minnesota    Mississippi       Missouri
##              2              1              2              1              1
##        Montana       Nebraska         Nevada  New Hampshire     New Jersey
##              2              2              1              2              2
##     New Mexico       New York North Carolina   North Dakota           Ohio
##              1              1              1              2              2
##       Oklahoma         Oregon   Pennsylvania   Rhode Island South Carolina
##              2              2              2              2              1
```

```
##   South Dakota      Tennessee          Texas           Utah        Vermont
##              2              1              1              2              2
##       Virginia     Washington  West Virginia      Wisconsin        Wyoming
##              2              2              2              2              2
## Objective function:
##    build     swap
## 1.441358 1.368969
##
## Available components:
##  [1] "medoids"    "id.med"     "clustering" "objective"  "isolation"
##  [6] "clusinfo"   "silinfo"    "diss"       "call"       "data"
```

The printed output shows:

- the cluster medoids: a matrix, which rows are the medoids and columns are variables.
- the clustering vector: A vector of integers (from 1:k) indicating the cluster to which each point is allocated

If you want to add the point classifications to the original data, use this:

```
dd <- cbind(USArrests, cluster = pam.res$cluster)
head(dd, n = 3)
```

```
##         Murder Assault UrbanPop Rape cluster
## Alabama   13.2     236       58 21.2       1
## Alaska    10.0     263       48 44.5       1
## Arizona    8.1     294       80 31.0       1
```

Accessing to the result of the pam() function

The function *pam()* returns an object of class pam which components include:

- medoids: Objects that represent clusters
- clustering: a vector containing the cluster number of each object

```
# Cluster medoids: New Mexico, Nebraska
pam.res$medoids
```

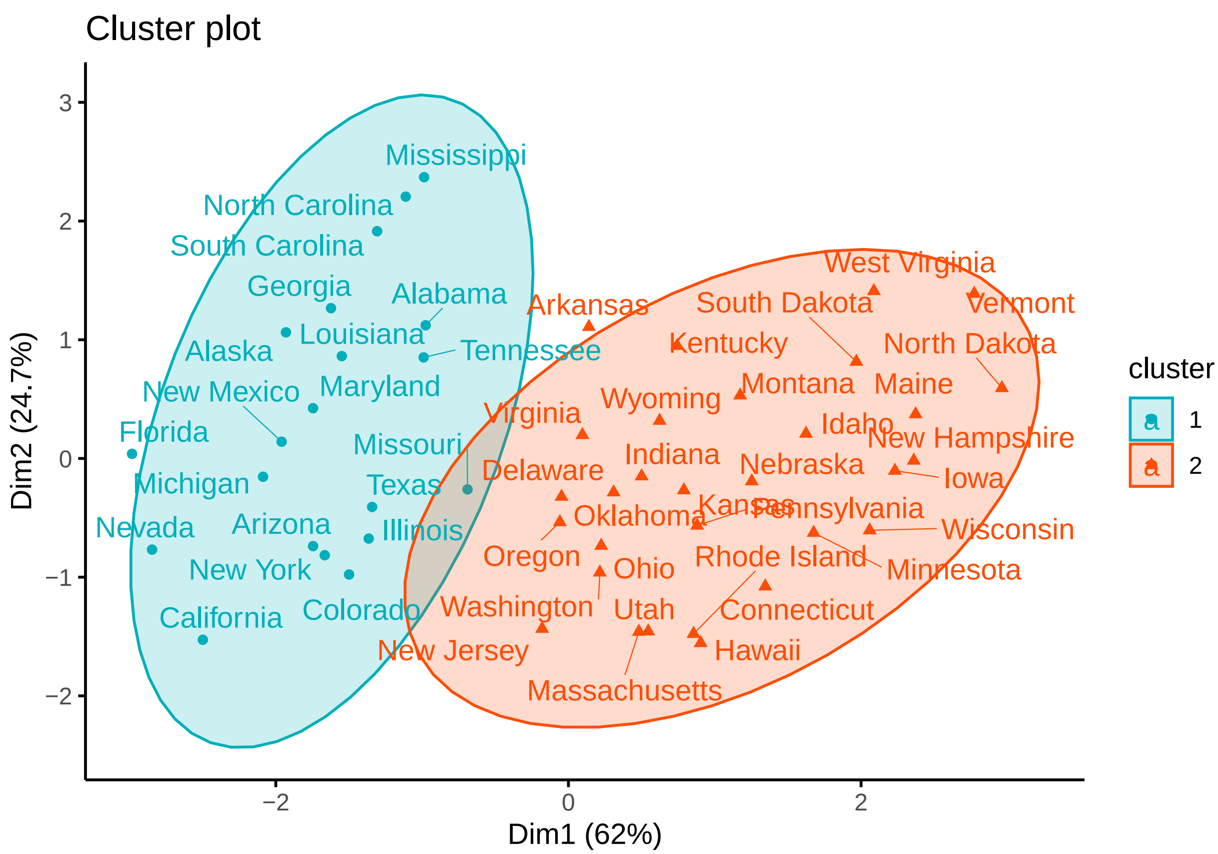
```
##                Murder    Assault   UrbanPop       Rape
## New Mexico  0.8292944  1.3708088  0.3081225  1.1603196
## Nebraska   -0.8008247 -0.8250772 -0.2445636 -0.5052109
```

```
# Cluster numbers
head(pam.res$clustering)
```

```
##    Alabama     Alaska    Arizona   Arkansas California   Colorado
##          1          1          1          2          1          1
```

Visualizing PAM clusters

```
fviz_cluster(pam.res,
             palette = c("#00AFBB", "#FC4E07"), # color palette
             ellipse.type = "t", # Concentration ellipse
             repel = TRUE, # Avoid label overplotting (slow)
             ggtheme = theme_classic()
             )
```



Note that, for large data sets, *pam*() may need too much memory or too much computation time. In this case, the function *clara*() is preferable. This should not be a problem for modern computers.